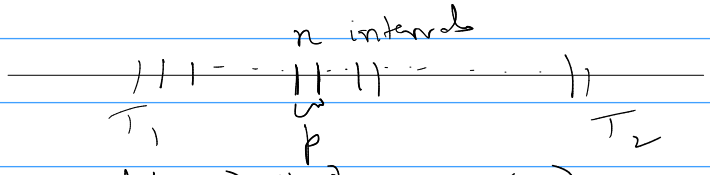
Sep 1 Poisson distribution $X \sim \text{Poisson}(\lambda)$ $(\lambda > 0)$

$$X \in \{0, 1, 2, \dots\}$$

$$p_X(k) = e^{-\lambda} \frac{\lambda^k}{k!}$$

Arrivals



λ : rate of arrival

$n \to \infty$

$np = \lambda$

$\text{Binomial}(n, p) \xrightarrow{np=\lambda} \text{Poisson}(\lambda)$

Continuous distributions

Σ_X: $\text{supp}(X) = [0, 1]$

→ assign probabilities to intervals

Σ_X: $X \in [a, b]$

Probability "density" function (PDF)

f_X : Function from $\mathbb{R} \to \mathbb{R}$ s.t.

$$\Pr(X \in [a, b]) = \int_a^b f_X(x)\, dx$$

Prob" for unit length"

→ $f_X(x) \geq 0$ in any interval

"almost everywhere"

$$\text{supp}(X) = \{x : f_X(x) > 0\}$$

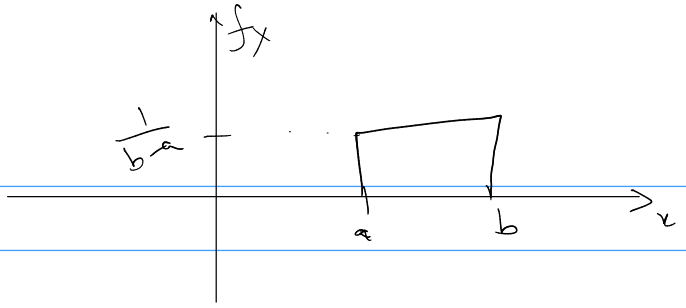
→ $$\int_{\text{supp}(X)} f_X(x)\, dx = 1$$

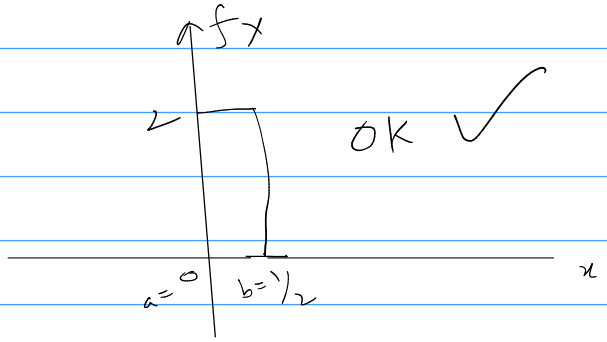
$$\int_{\mathbb{R}} f_X(x)\, dx = \int_{\text{supp}(X)} f_X(x)\, dx$$

Σ_X: ① Uniform distribution

$X \sim \text{Uniform}[a, b]$

$$f_X(x) = \begin{cases} \frac{1}{b-a}, & a \leq x \leq b \\ 0, & \text{else} \end{cases}$$





Probability of $[a, a]$: $\int_a^a f_X(x)\,dx = 0$

$Pr(X \in [a,b]) = Pr(X \in (a,b]) = Pr(X \in (a,b)) = Pr(X \in [a,b))$

Ex: ① $f_X(x) \propto x$ in $[0, 10]$.

$f_X(x) = cx$ $\int_0^{10} cx\,dx = 1$ $c = \frac{1}{50}$

② $f_X(x) \propto x(1-x)$ in $[0,1]$

$\int f_X(x)\,dx$ — $\frac{}{dx}$

③ $f_X(x) \propto e^{-\lambda x}$ in $[0, \infty]$

$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x > 0 \\ 0, & \text{else} \end{cases}$

$Pr(X \geq a) = e^{-\lambda a}$

$Pr(a < X < b) = e^{-\lambda a} - e^{-\lambda b}$ ($a, b > 0$, $a < b$)

④ $f_X(x) \propto e^{-x^2/2}$ in $\mathbb{R}$

$f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$, $x \in \mathbb{R}$

Cumulative Distribution Function (CDF)

(S, Events, P)

RV $X : S \to \mathbb{R}$

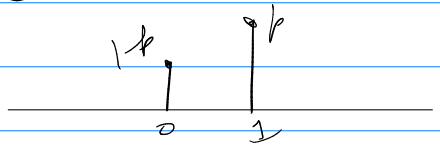
→ CDF of X, denoted F_X, is a function from $\mathbb{R}$ to $[0,1]$

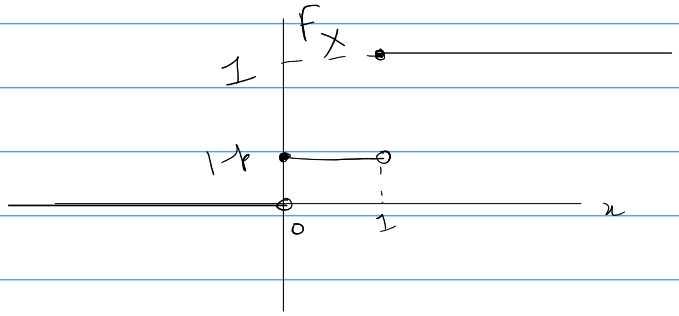
and is defined as

$$F_X(x) = P(X \le x)$$

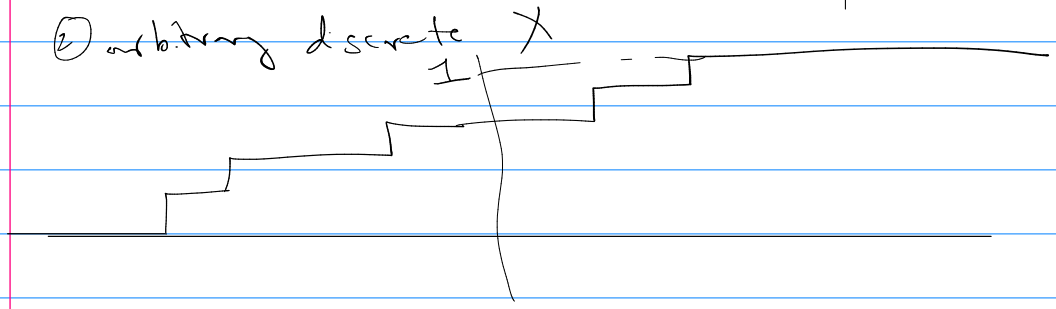
$$\text{Event}: \{\omega \in S : X(\omega) \le x\}$$

Ex: ① $X \sim \text{Bernoulli}(p)$





② arbitrary discrete X



③ $X \sim \text{Uniform}[a, b]$

